Design of Enterprise IoT: Assignment 1a

Smart Turtle Breeding System (STBS)

Team 01

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1 Task 1: Use Case Description

A Chinese soft-shell turtle farm operates 20 indoor or semi-closed aquaculture units with an operations and maintenance team. Farm operators manage water quality, feeding, temperature, turtle activity and nesting areas; maintenance technicians service filters and pumps.

Operators manually measure water conditions, prepare feed, inspect behaviour and health, change water, and search for eggs. These checks are time-consuming, problems may go unnoticed between rounds, and feeding times and amounts may vary. Periodic rounds can therefore cause inconsistent feeding, delayed detection of deteriorating conditions or equipment failures, and labour-intensive egg collection. Water quality, feeding and temperature affect turtle health and growth [1]. Reviews identify temperature, dissolved oxygen and pH as priority IoT aquaculture parameters [2, 3].

An IoT solution is proposed to monitor water quality and turtle behaviour continuously, control feeding and temperature, and support bounded partial water changes. Distributed sensors, actuators and cameras send alerts and records so one team can respond between rounds. The expected benefits are less routine work, earlier warnings and more consistent feeding; field evidence also shows that vision-based nesting detection can reduce egg-harvesting effort [4].

2 Task 2: Solution Overview

STBS uses water-temperature, pH, turbidity and ammonia sensors in each unit, with level/flow feedback, feeder sensors and IP cameras covering activity, feeding and nesting zones. Local controllers compare readings with species- and growth-stage set-points, regulate heaters, dispense scheduled portions and perform bounded partial water changes. Device feedback and follow-up measurements verify each

action. An edge gateway keeps essential rules operating during internet loss, filters routine data and sends readings, events and selected video to a cloud dashboard. Alerts identify the affected unit, abnormal parameter, action taken and need for intervention. Farm operators configure targets, review alerts and nesting evidence, collect eggs and override automation. Maintenance technicians calibrate sensors, service filters and equipment, diagnose failures and maintain power/network availability. Role-based access, audit logs and local fail-safe limits support reliable operation.

3 Task 3: Value Proposition

The value proposition distinguishes the current substitute from the proposed system, separates benefits from features, describes the user experience, and records adoption needs and implementation fears. Each statement identifies the stakeholder to whom it applies.

Table 1: STBS value proposition.

Substitute	**Benefits**
• Farm operators have to manually measure quality and temperature of the water in each breeding unit. (Farm Operators)	• The system can help maintain suitable water conditions and reduce the duration of abnormalities. (Farm Operators)
• Farm operators manually feed turtles stick to daily schedules. (Farm Operators)	• The system can improve feeding consistency and reduce missed or inaccurate feeding. (Farm Operators)
• Farm operators have to regularly inspect turtle activity, feeding behaviour, and health conditions. (Farm Operators)	• The system can help detect suspected abnormal behaviour earlier, allowing timely inspections. (Farm Operators)
• Farm operators manually inspect nesting areas and collect turtle eggs. (Farm Operators)	• The system can identify suspected egg-laying activities and locations for on-site verification. (Farm Operators)
• Farm operators manually change water, while maintenance staff schedule filter maintenance based on experience and inspection records. (Farm Operators and Maintenance Technicians)	• The system can reduce repetitive work and support condition-based filter maintenance using sensor data and operational records. (Farm Operators and Maintenance Technicians)
Features	**Experience**
• The system continuously monitors water temperature, pH, turbidity, and ammonia nitrogen levels, and automatically regulates water temperature. (Farm Operators)	• Farm operators feel more confident as they can monitor all 20 breeding units through a centralised dashboard. (Farm Operators)
• The system automatically dispenses scheduled feed portions and verifies delivery through sensor feedback. (Farm Operators)	• Farm operators feel less worried about unnoticed abnormalities, as they receive alerts with abnormal readings and corrective actions. (Farm Operators)

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Substitute	Benefits
• The system uses video analytics to detect turtle activity, feeding behaviour, and suspected nesting events, with recordings for verification. (Farm Operators)	• Farm operators feel more in control as they can track automated operations and intervene when necessary. (Farm Operators)
• The system automatically controls pumps and valves for partial water changes, verifying water levels, flow rates, and water quality afterwards. (Farm Operators and Maintenance Technicians)	• Farm operators feel less burdened by repeated nesting inspections, as they can review videos and records remotely. (Farm Operators)
• The system connects local controllers to the cloud through a gateway, enabling remote monitoring, alerts, operation verification, and manual override. (Farm Operators and Maintenance Technicians)	
Fears	**Needs**
• Power or network failures may interrupt monitoring, automated control, or remote alerts. (Farm Operators and Maintenance Technicians)	• Farm operators need to maintain suitable water conditions for different turtle species and growth stages. (Farm Operators)
• Sensor fouling, ageing, or calibration errors may lead to inaccurate readings and control decisions. (Farm Operators and Maintenance Technicians)	• Farm operators need to reduce repetitive tasks and manage all 20 breeding units more efficiently. (Farm Operators)
• Video analytics may falsely identify or miss abnormal behaviour and nesting events, requiring on-site verification. (Farm Operators)	• Farm operators need remote monitoring, timely alerts, and verification of automated corrective actions. (Farm Operators)
• Failures of feeders, heaters, pumps, or valves may disrupt operations and breeding conditions. (Farm Operators and Maintenance Technicians)	• Farm operators need reliable behaviour and nesting records to support inspections and breeding management. (Farm Operators)
• High installation costs and ongoing expenses for electricity, calibration, repairs, and consumables may increase the financial burden. (Farm Operators)	• The farm needs an affordable, easy-to-maintain system with manual override capabilities and maintenance records. (Farm Operators and Maintenance Technicians)

4 References

[1] Food and Agriculture Organization of the United Nations. "Cultured Aquatic Species Information Programme: Chinese soft-shell turtle (*Trionyx sinensis*)." Available at FAO cultured species profile.

[2] D. R. Prapti et al. "Internet of Things (IoT)-based aquaculture: An overview of IoT application on water quality monitoring." *Reviews in Aquaculture*, 14, 979–992, 2022. doi:10.1111/raq.12637.

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[4] A. Munyaev, B.-L. Jiang, and I.-J. Huang. “Smart turtle farm: Soft-shell turtle nesting behavior recognition and monitoring system.” *Computers and Electronics in Agriculture*, 244, 111477, 2026. doi:10.1016/j.compag.2026.111477.